

SUPERSTORE SALES DASHBOARD

Business Insights Report — Based on the Submitted Power BI Video

Purpose: Document the dashboard, its interactive behavior, KPI results, business insights, DAX measures and recommendations as demonstrated in the submitted screen recording. The filter results below are transcribed from the demonstrated video states.

1. Dashboard Overview

The dashboard is an interactive SuperStore Sales Dashboard. It uses Year, Segment, Region and Ship Mode as slicers. The main KPI cards are Sales, Profit, Profit Margin, Quantity and AOV. The page also contains a monthly Sales/Profit line chart, Profit by Category donut chart, State/Region map, Sales by City bar chart and Sales by Sub-Category bar chart.

2. Initial Dashboard Performance

KPI	Visible Result	Interpretation
Total Sales	632.63K	Overall sales shown in the initial dashboard state.
Total Profit	85.80K	Overall profit shown in the initial dashboard state.
Profit Margin	11.68%	Profit as a percentage of sales.
Quantity	10K	Total quantity displayed on the dashboard.
AOV	466	Average Order Value displayed by the dashboard.

3. Interactive Filter Demonstrations in the Video

This section clearly separates every filter selection into columns. Each row represents one demonstrated interaction in the video and records the resulting KPI values.

Time	Year	Segment	Region	Ship Mode	Sales	Profit	Profit Margin	Quantity	AOV
~20 sec	2014	Consumer	East	Same Day	401.21	-6	-1.51%	24	466
~40 sec	2017	Corporate	South	Second Class	3.61K	0.73K	20.36%	39	466
~60 sec	All	Home Office	West	First Class	12.48K	714.33	12.20%	157	466
~80 sec	All	All	West	First Class	11.84K	2.36K	29.27%	112	466
~100 sec	All	All	All	First Class	115.93K	14.23K	11.33%	2K	466

Dashboard response after each filter selection

Time	Dashboard Response
~20 sec	The dashboard updates to the 2014 + Consumer + East + Same Day subset. KPI cards and the category, city and sub-category visuals change to the filtered data.
~40 sec	The dashboard updates to 2017 + Corporate + South + Second Class. The monthly trend, regional map and ranking visuals refresh.
~60 sec	The dashboard updates to All Years + Home Office + West + First Class. West-focused geography and new city/sub-category rankings are displayed.
~80 sec	Segment is changed to All while West + First Class remain selected. KPI values and visual rankings update to the broader selection.
~100 sec	Region is changed to All while First Class remains selected. The map broadens and the city/sub-category analysis updates.

4. Business Insights from the Dashboard and Video

- **Interactive dashboard behavior:** Year, Segment, Region and Ship Mode filters dynamically change the KPI cards and visual analysis.
- **Profitability varies by selection:** the demonstrated combinations show profit margin values from **-1.51%** to **29.27%**, indicating that filter context materially affects profitability.
- **Initial category profitability:** Technology represents **59.15%** of visible profit, Office Supplies **31.49%**, and Furniture **9.35%**.
- **Initial city performance:** New York is the highest displayed city at approximately **74K**, followed by Los Angeles (49K), San Francisco (31K), Seattle (25K), and Philadelphia (23K).
- **Initial sub-category performance:** Chairs leads the displayed ranking at approximately **91K**, followed by Phones (84K), Machines (74K), Storage (55K), and Tables (54K).
- **Monthly trend:** the Sales and Profit line chart changes with the selected filters, allowing month-level performance to be analyzed in context.
- **AOV:** the video displays **466** across the demonstrated filter states. This should be checked against the final PBIX AOV definition before submission.

5. Business Recommendations

- Focus on Technology for profit-oriented growth because it has the largest visible profit contribution.
- Review Furniture profitability and identify whether pricing, discounting or product mix is limiting its contribution.
- Use city-level analysis to prioritize high-performing markets such as New York.
- Monitor Chairs and Phones as key sales-driving sub-categories.
- Use filter combinations to compare customer segments, regions and shipping modes before making operational decisions.
- Verify the AOV calculation and currency formatting in the final Power BI file.

6. Power BI DAX Measures

Measure	DAX Formula	
Total Sales		Total Sales = SUM('Superstore'[Sales])
Total Profit		Total Profit = SUM('Superstore'[Profit])
Total Quantity		Total Quantity = SUM('Superstore'[Quantity])
Total Orders		Total Orders = DISTINCTCOUNT('Superstore'[Order ID])
Profit Margin %		Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
Average Order Value		AOV = DIVIDE([Total Sales], [Total Orders], 0)

7. Recommended Date Table DAX

```
DateTable =  
ADDCOLUMNS(
```

```
CALENDAR(
MIN('Superstore'[Order Date]),
MAX('Superstore'[Order Date])
),
"Year", YEAR([Date]),
"Month Number", MONTH([Date]),
"Month", FORMAT([Date], "MMMM"),
"Month Short", FORMAT([Date], "MMM"),
"Month Year", FORMAT([Date], "MMM YYYY"),
"Quarter", "Q" & FORMAT([Date], "Q")
)
```

Create a relationship between DataTable[Date] and Superstore[Order Date]. Sort Month and Month Short by Month Number for correct January–December ordering.

8. Dashboard Visual-to-Measure Mapping

Dashboard Visual	Fields / Measures
Sales KPI	[Total Sales]
Profit KPI	[Total Profit]
Profit Margin KPI	[Profit Margin %]
Quantity KPI	[Total Quantity]
AOV KPI	[AOV]
Monthly Sales & Profit	Month + [Total Sales] + [Total Profit]
Profit by Category	Category + [Total Profit]
Sales by State and Region	State + Region + [Total Sales]
Sales by City	City + [Total Sales]
Sales by Sub-Category	Sub-Category + [Total Sales]

9. Final Submission Checklist

- Power BI dashboard with working Year, Segment, Region and Ship Mode slicers.
- Business Insights PDF — this report.
- README documentation.
- Public GitHub repository containing project files/source materials.
- Verify AOV DAX and currency formatting in the PBIX.
- Replace the legacy/retiring map visual shown in the recording with a currently supported map visual before final submission.

Assignment alignment: Task 1 requires an interactive sales dashboard covering data cleaning/preprocessing, sales KPI analysis, interactive charts, regional analysis, product performance and business insights, with a Business Insights PDF as a deliverable.